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Notes:

- **To obtain full marks for each question, relevant and clear steps need to be included in the answers.**
- **Partial marks may be awarded depending on the degree of completeness and clarity.**

Question 1: Proof Techniques **[8 marks]**

(a) Use the proof by contradiction method to verify the following statement: if a and b are rational numbers and $b \neq 0$, then $a + \sqrt{3}b$ is irrational.

(4 marks)

(b) Use the proof by induction method to verify the following statement given that n is a natural number and $n \geq 1$,

$$2 \cdot 2^1 + 3 \cdot 2^2 + 4 \cdot 2^3 + \ldots + (n+1) \cdot 2^n = n \cdot 2^{n+1}$$

(4 marks)

Question 2: Set Theory **[12 marks]**

(a) Let the universal set be $\mathbb{Z}$, $P = \{p \in \mathbb{Z} \mid p \text{ is odd and } 1 < p < 9\}$,
$Q = \{q \in \mathbb{Z} \mid q \text{ is even}, q \neq 0, q \neq 4 \text{ and } 3 < q < 10\}$,
$R = \{r \in \mathbb{Z} \mid r \text{ is odd}, 1 < r < 17 \text{ and } 6r = 90\}$ and $T = \{t \in \mathbb{Z} \mid t^2 + 3t - 40 = 0\}$.

Find the elements of the following expressions:

1. $T \cap (P \cup R)$
2. $(R - P) \cup Q$
3. $(P \Delta R) \cap (Q \cap \sim P)$
4. $((P \cup R) \cap Q) - ((Q \cap \sim P) \cap (P \Delta R))$

(4 marks)

(b) Let M and N be sets, then $M \subseteq N$ if and only if $\sim M \subseteq \sim N$. Prove or disprove it.

(8 marks)

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Question 3: Relations **[20 marks]**

(a) Let P = {5,4,3,2,1} and Q ={(1,2),(2,2),(3,1),(4,5),(5,2),(4,3),(4,4),(5,4)}. Firstly represent Q over P using a digraph, then determine the corresponding adjacency matrix of the digraph.

(4 marks)

(b) If M and N are both transitive relations, then their intersection is also transitive. If you think that it is true, prove it. If not, give a counter-example.

(8 marks)

(c) Define S be a relation on $\mathbb{N}$ denoted as xSy if $x^2 + y^2$ is even. If you think that S is an equivalence relation, prove it. If not, give a counterexample.

(8 marks)

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Question 4: Functions **[18 marks]**

(a) Let $f: \mathbb{R} \to \mathbb{R}$, check whether $f(y) = \frac{y+7}{3y-4}$ and $f^{-1}(y) = \frac{4y+7}{3y-1}$ are inverses.

(4 marks)

(b) Let A, B and C be non-empty finite sets, and $f: A \to B$, $g: B \to C$ and $h: A \to C$. Prove or disprove the following:

1. If f is surjective and g is not injective, then $g \circ f$ is not injective.
2. If f and g are bijective, and h is defined by h(x) = f(g(x)), then h is bijective.

(10 marks)

(c) Let $f: \mathbb{R} \to \mathbb{R}$ be given by $f(x) = 5x^2 - 7$ and $g: \mathbb{R} \to \mathbb{R}$ be given by $g(x) = 8x + 2$. Find $(f \circ g)(x)$, $(g \circ f)(x)$, $(f \circ f)(x)$ and $(g \circ g)(x)$.

(4 marks)

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Question 5: Logic **[22 marks]**

(a) For each of the two cases 1. and 2. below, show whether the statement on the left-hand side is equivalent to the statement on the right-hand side:

1. $(p \wedge q) \vee (\neg p \wedge \neg q) \equiv p \Leftrightarrow q$
2. $\neg((p \rightarrow q) \wedge (q \rightarrow p)) \equiv p \Leftrightarrow \neg q$

(8 marks)

(b) Let p, q and r stand for the following propositions:

- p: "Adam feels hungry."
- q: "Adam eats many hot dogs."
- r: "Adam's mother is happy."

Use p, q and r to translate the following into propositional logic:

1. If Adam feels hungry and eats many hot dogs, then Adam's mother is not happy.
2. Adam eats many hot dogs only if he feels hungry.

(4 marks)

(c) Let S be the signature consisting of a unary predicate symbol *Book*, a binary predicate symbol *IsWriterOf* and two individual constants *dickens* and *oliver*. Assume that these symbols have the following meaning:

- *Book(x)* states: x is a book.
- *IsWriterOf(x, y)* states: x is a writer of y.
- *dickens* and *olivertwist* refer to "Charles Dickens" and "Oliver Twist", respectively.

Translate the following sentences into S-formulae:

1. Oliver Twist is not a book.
2. Charles Dickens is a writer of Oliver Twist.
3. There is a book that Charles Dickens is a writer of.
4. For every book there is a writer.
5. There is a book whose only writer is Charles Dickens.

(10 marks)

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Question 6: Combinatorics and Probability **[20 marks]**

(a) Let X = {1, 2, 3, 4, 5, 6, 7, 8}. Show that if five distinct integers are randomly selected from X, then at least one pair of integers will have a sum of 9.

(3 marks)

(b) Three fair coins are flipped. Let A be the event that all heads or all tails come up, B be the event that at least 2 heads come up, and C be the event that not all tails come up. Consider the following pairs of events: (A, B), (A, C) and (B, C).
Determine which pairs are independent and which pairs are dependent.

(4 marks)

(c) There are 400 dormitory rooms along with 400 distinct room numbers for 60 CPT107 students. Any CPT107 student can stay in any of the 400 dormitory rooms.

1. Suppose you are a CPT107 student and are recording the room number for all CPT107 students, one at a time until you have found a match (this means that a room number has already been recorded).
   i. What is the probability that it takes more than 30 students for this to happen?
   ii. What is the probability that it takes exactly 25 people for this to happen?
2. Suppose you are a CPT107 student and are recording the room number for all CPT107 students, one at a time until you have found a student who shares the same dormitory room with you. What is the probability that it takes exactly 28 CPT107 students for this to happen?

(9 marks)

(d) There are 40 balls in a box, of which 35 of them are black and the rest are white. A player has to pay \$100 to pick a ball randomly from the box. If the player gets a white ball, he wins \$750. What is the expected profit of this bet?

(4 marks)

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